NNN-C-940B
March 10, 1970
SUPERSEDING
Int. Fed. Spec. NNN-C-00940A(GSA-FSS)
July 11, 1968 and
Fed. Spec. NNN-C-940
March 16, 1963
(See 6.4)

FEDERAL SPECIFICATION

CYLINDER, GRADUATED, LABORATORY; GLASS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers glass graduated cylinders suitable for laboratory purposes.

1.2 Classification.

1.2.1 Styles and sizes. Graduated cylinders covered by this specification shall be of the following styles and sizes, as specified (see 6.2):

Styles and sizes.

Style I - With pour-out lip.
- Size 1 - 5 milliliters (ml.).
- Size 2 - 10 ml. (intermediate graduations).
- Size 3 - 10 ml. (no intermediate graduations).
- Size 4 - 25 ml. (2 ml. main graduations).
- Size 5 - 25 ml. (5 ml. main graduations).
- Size 6 - 50 ml.
- Size 7 - 100 ml.
- Size 8 - 250 ml.
- Size 9 - 500 ml.
- Size 10 - 1000 ml.
- Size 11 - 2000 ml.

Style II - With standard-taper stopper.
- Size 1 - 10 ml. (intermediate graduations).
- Size 2 - 10 ml. (no intermediate graduations).
- Size 3 - 25 ml. (2 ml. main graduations).
- Size 4 - 25 ml. (5 ml. main graduations).
- Size 5 - 50 ml.
- Size 6 - 100 ml.
- Size 7 - 250 ml.
- Size 8 - 500 ml.
- Size 9 - 1000 ml.
- Size 10 - 2000 ml.

Style III - With pour-out lip and reinforced top.
- Size 1 - 100 ml.
- Size 2 - 250 ml.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

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Federal Specifications:

DD-G-541 - Glass (Laboratory).
PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-B-676 - Boxes, Set-Up.
PPP-C-843 - Cushioning, Material, Cellulosic.
PPP-T-45 - Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, Washington.

(Federal Government activities may obtain copies of Federal specifications, standards, and handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids shall apply.

National Classification Board:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Tariff Order Section, 1616 P Street, N. W., Washington, D. C. 20036.)

Uniform Classification Committee:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

Commercial Standard:

CS21 - Interchangeable Taper-Ground Joints, Stopcocks, Stoppers, and Spherical-Ground Joints.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402.)

American Society for Testing and Materials (ASTM) Standards:

C 148-65 - Polariscopic Examination of Glass Containers.
C 162-66 - Definitions of Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government 1916 Race Street, Philadelphia, Pa., 19103.)

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3. REQUIREMENTS

3.1 Material. Graduated cylinders shall be of borosilicate glass conforming to the requirements of type I, class A of DD-G-541.

3.2 Annealing. Ring strain shall be no greater than Temper No. 1 when tested as specified in 4.4.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.3 Capacity accuracy. Capacity accuracy of all graduated cylinders shall be in accordance with the tolerances of table I when tested as specified in 4.4.2. The graduated cylinders shall be calibrated at 20° C. Style I shall be calibrated to deliver or to contain; style II and style III shall be calibrated to contain (see 6.2).

3.4 Design.

3.4.1 Shape. Graduated cylinders shall be of one piece, consisting in general of a top, having either a pour-out lip (style I or style III) or a standard taper ground glass stopper (style II); a graduated portion of uniform diameter; and a base. The height of the graduated portion shall be at least 5 times the inside diameter except for sizes 3 and 5 of style I and sizes 2 and 4 of style II where the height of the graduated portion shall be at least 4 times the inside diameter. Any cross section of the graduated portion taken in a plane perpendicular to the vertical axis shall be circular. The shape shall permit complete emptying and thorough cleaning. Style III shall have an integral glass bead blown around the outside of the cylinder near and under the pour-out lip. The lips of all three styles shall be located in a vertical line with an angle on the hexagonal base and that of style III shall not extend beyond the bead. Construction shall be such that upon falling the bead shall strike a flat surface before the lip.

3.4.2 Base. The base shall be perpendicular to the vertical axis of the graduated cylinder. The construction of the base shall be such that the empty graduated cylinder (without stopper for style II) in capacities above 10 ml. will stand on a surface inclined 15 degrees with the horizontal and in capacities of 10 ml. or less will stand on a surface inclined 10 degrees with the horizontal. Bases shall be hexagonal.

3.4.3 Dimensions. The dimensions and tolerances shall be as shown in table I. The glass stoppers of style II graduated cylinders shall conform to Commercial Standard CS21.

3.4.4 Graduation markings. Graduation markings shall be sharply defined lines of uniform width not less than 0.2 millimeters (mm.). Graduation markings shall be perpendicular to the vertical axis of the cylinder and parallel to each other. Adjacent graduation markings shall be spaced at least 1 mm. from center to center. Markings shall be applied by one of the following methods - etched and filled with a permanent pigment; etched through a vertical colored band fired into the glass; or by application of a stain which is fired into the glass without etching; or by application of enamel fired into the glass without etching; or by application of an enamel which is fired onto a vertical colored etched band which has been fired into the glass. Main graduations shall extend at least 3/4 of the way around the cylinder; intermediate graduations shall extend at least 1/5 of the way around the cylinder; and least graduations shall extend at least 1/7 of the way around the cylinder. The value of all main graduations shall be suitably inscribed in Arabic numerals directly above the line referenced. The markings shall not change appreciably in appearance when tested as specified in 4.4.3. Subdivision lines may be omitted between the base of the cylinder and the first numbered line. Style III shall have graduation markings etched through a vertical red colored band fired into the glass.

3.5 Identification marking. Each graduated cylinder shall be permanently and legibly marked with the manufacturer's name or trademark, the capacity, and the temperature of calibration (20° C.). Style I shall be marked to deliver (TD) or to contain (TC); style II and style III shall be marked to contain (TC).

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TABLE I. Dimensions and tolerances for graduated cylinders

Capacity, ml.	Applicable size when Style I	Style II	Style III	Graduations ml. Main	Intermediate	Least	Distance from scale to top, cm. Maximum	Minimum*	Minimum wall thickness, mm.	Stopper No. **	Plus or minus tolerance ml.
5	1	--	--	1.0	0.5	0.1	5	2	0.9	9	0.1
10	2	1	--	1.0	0.5	0.1	5	2	1.0	9	0.1
10	3	2	--	1.0	---	0.2	5	2	1.0	13	0.1
25	4	3	--	2.0	1.0	0.2	8	2	1.2	13	0.3
25	5	4	--	5.0	1.0	0.5	5	2	1.2	13	0.3
***50	6	5	--	5.0 or 10.0	5.0	1.0	8	2.5	1.2	16	0.4
100	7	6	1	10.0	5.0	1.0	7	3	1.5	16 or 22	0.6
250	8	7	2	20.0	10.0	2.0	8	4	1.5	22 or 27	1.4
500	9	8	--	50.0	25.0	5.0	10	4.5	1.7	27 or 32	2.6
1000	10	9	--	100.0	50.0	10.0	10	4.5	2.0	32	5.0
2000	11	10	--	200.0	100.0	20.0	10	5	2.0	38	10.0

*For style II, measurement is to bottom of glass stopper.
**Applicable to style II only.
***For 50 ml. cylinder, main graduations may be each 5 or 10 ml.; if main graduations are each 5 ml.; intermediate graduations shall not be applicable.

3.6 Workmanship. All graduated cylinders shall be transparent and free from chips, cracks, stones, and other defects which detract from their appearance or distort the appearance of the liquid surface under the graduated portion, or which may impair the graduated cylinders serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification is defined as both examination (such as visual, tactile, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificate of analysis. Certified test data, in lieu of contractor performing tests, will be acceptable as evidence of compliance with requirements of 3.1.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein. For purposes of sampling, an inspection lot for acceptance inspection and tests shall consist of all material of the type submitted for inspection and delivery at one time.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Unit of product shall be the cylinder.

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Characteristic	Requirement	Test procedure	Inspection level
Strain test	3.2	4.4.1	S-2
Capacity accuracy	3.3	4.4.2	S-3
Permanency of marking	3.4.4	4.4.3	S-1

4.3 Examination. The cylinders shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories	Defects*
Major	
101	Cylinder not free of cracks.
102	Cylinder not free of chips.
103	Cylinder not free of sharp edges.
104	Cylinder not free of protrusions.
105	Cylinder not free of deformation.
106	Graduation and calibration markings not complete, correct, or legible.
Minor	
201	Manufacturer's identification not complete, correct, or legible.
202	Cylinder not free of bubbles.
203	Cylinder not free of stones and striae.

*Examination shall not be restricted to the classified possible defects listed above. Definitions of glass defects shall be in accordance with ASTM C 162-66.

4.3.2 Dimensional examination. The cylinders shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Level A and B preparation for delivery (section 5), inspection level II.

Categories	Defects
Critical:	None defined.
Major	AQL 1.5 percent defective.
101	Unit, intermediate, or shipping containers not as specified.
102	Number of cylinders per intermediate or shipping container not as specified or indicated.
103	Unlike cylinders packaged or packed in same containers.
104	Unit, intermediate, or shipping containers damaged.
105	Unit, intermediate, or shipping container closure not as specified.
106	Marking of unit, intermediate, or shipping container illegible, incorrect, or incomplete.

4.4 Tests. Distilled water and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.4.1 Annealing. Graduated cylinders shall be tested in accordance with method A or B of ASTM C 148-65.

4.4.2 Marking accuracy. After thorough cleaning, the graduated cylinder shall be filled accurately with water to the graduation mark under test using calibrated pipets or a buret. Three main graduations (including the capacity graduation mark) at approximately equal intervals shall be tested.

4.4.3 Permanency of marking. A different graduated cylinder shall be used for each of the following tests.

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4.4.3.1 Resistance to alkali. The graduated cylinder shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for one half hour, and allowed to stand in the alkali for an additional one hour. The graduated cylinder shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.3.2 Resistance to acid.

4.4.3.2.1 Hydrochloric acid. The graduated cylinder shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for one half hour, and allowed to stand in the acid for an additional one hour. The graduated cylinder shall then be removed thoroughly rinsed with distilled water and dried by rubbing vigorously with a cloth.

4.4.3.2.2 Sulfuric acid. The graduated cylinder shall be completely immersed in concentrated sulfuric acid (specific gravity 1.84) at room temperature and allowed to stand one hour. The graduated cylinder shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.5 Acceptance/rejection criteria. If any graduated cylinder fails to meet the test requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each graduated cylinder shall be packaged in a box conforming to PPP-B-566, style I or II, type and class optional; PPP-B-636, type CF, class domestic, or PPP-B-676, type, class and style optional. Graduated cylinders, provided with glass stoppers, shall have the stopper secured in the box. The box shall be provided with fiberboard blocking or sufficient cushioning material conforming to type II, class B of PPP-C-843 to insure against shifting of or damage to the graduated cylinder. Box closure shall be secured with tape conforming to PPP-T-45.

5.1.1.2 Intermediate packaging. The number of unit packages specified (see 6.2) shall be intermediately packaged in a box conforming to PPP-B-636, grade W5c. Each intermediate package shall contain only one style, size, and type graduated cylinder. The boxes shall be closed in accordance with the appendix of the box specification.

5.1.1.3 Level C. Graduated cylinders shall be packaged in accordance with the manufacturer's commercial practice, suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Graduated cylinders shall be packed in a shipping container conforming to PPP-B-621, class 2 or PPP-B-601, overseas type. As far as practical, exterior containers shall be of uniform size and shape, be of minimum cube and tare, and contain identical quantities of like items. Gross weight of mailed wood boxes shall not exceed 200 pounds. Containers shall be closed and strapped in accordance with the applicable box specification or appendix thereto.

5.2.2 Level B.

5.2.2.1 Civil agencies. Graduated cylinders in the quantity specified (see 6.2), packaged as specified in 5.1, shall be packed in a fiberboard box conforming to PPP-B-636, class domestic, grade not less than 275.

5.2.2.2 Military agencies. Graduated cylinders shall be packed in accordance with 5.2.1 except that shipping containers shall conform to PPP-B-621, class 1 or PPP-B-601, domestic type.

5.2.3 Level C. Graduated cylinders shall be packed to insure carrier acceptance and safe delivery at first domestic destination. Containers shall be in accordance with Uniform Freight Classification Rules and National Motor Freight Classification Rules as applicable to the mode of transportation.

5.3 Marking.

5.3.1 Civil agencies. In addition to any special marking required by the contract or order, marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. Graduated cylinders covered by this specification are intended for laboratory applications.

6.2 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents.

(a) Title, number, and date of this specification.
(b) Level of packaging and packing required(see 5.1 and 5.2).
(c) Style and size required (see 1.2.1).
(d) Number of cylinders to intermediate package and shipping container (see 5.1.1.2 and 5.2.2.1).
(e) Cylinders calibrated "to contain" or "to deliver", if required (see 3.3).
(f) Special markings required (see 5.3.1 and 5.3.2).

6.3 Military use. Items procured under this specification for military use are to be limited to the variety shown on the applicable military standard or military supply standard. Personnel of the military departments are requested to refer to these documents for guidance.

6.4 Supersession data. This specification supersedes Interim Federal Specification NNN-C-00940A(GSA-FSS) and Federal Specification NNN-C-940. Federal Specification NNN-C-940 superseded in part Federal Specification DD-V-581A; in particular, the requirements applicable to type V, graduated cylinders.

MILITARY INTEREST:

Custodians:

Army - MU
Navy - MS

Review activities:

Army - MD
Navy - MS
DSA - DP

User activities:

Army - MI, SM

Preparing activity:

GSA-FSS

CIVIL AGENCIES INTEREST:

AGR
COM-NBS
GSA-FSS
HEW
VA-DMS

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GENERAL SERVICES ADMINISTRATION - FEDERAL SUPPLY SERVICE

SPECIFICATION COMMENT SHEET

BUDGET BUREAU NO. 29-R0175

INSTRUCTIONS

This form provides a way for users of this specification to inform the originator of problems encountered in its use. It is not to be used to request changes to accommodate proprietary features. All comments will be considered and appreciated, but please do not expect a reply. To comment: detach, complete, and mail to General Services Adm., FSS (FMSC), Washington, D. C. 20406.

NOTE: Comments on this form do not constitute or imply authorization to waive any part of the document or serve to amend contractual requirements.

1. SPECIFICATION

NNN-C-940B CYLINDER, GRADUATED, LABORATORY; GLASS

2. CONTRACT NO. *(If any)*	3. QUANTITY ON CONTRACT *(Optional)*	4. DOLLAR VALUE *(Optional)*

5. GENERAL NATURE OF PROBLEM *(e.g., inspection difficulties, manufacturers unable to meet tolerances, containers collapse under normal warehousing conditions, etc.)*

6. SPECIFIC REQUIREMENTS AFFECTED *(Include paragraph number and lines of wording)*

7. SPECIFIC PROBLEMS *(e.g. tests in 4.2.2 will not assure that the battery will last required time; temperature ranges in table 2 do not conform to commercially available items.)*

8. RECOMMENDATIONS

9. NAME OF MANUFACTURER, ASSOCIATION, GOVT., AGENCY, ETC.	10. ADDRESS *(Number, Street, City, State and Zip Code)*

11. NAME AND TITLE OF SUBMITTER	12. DATE

GSA DC 69-9761

FORM GSA FEB 69 2200

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